Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Full-stack software engineer with 6+ years of deep domain expertise in affordable housing compliance software. Spent 6 years at Entrata building the systems that property managers rely on daily for LIHTC certification, HUD income-limit compliance, TRACS/iMAX submissions, and resident eligibility workflows -- across a 100K+ MAU multi-tenant SaaS platform. Owned the HUD income-limit import end-to-end for 6 years with zero compliance violations. Currently at Intuit building Python/FastAPI services on AWS. Strong in React, Python, SQL, AWS, agile delivery, TDD, and code review. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Affordable Housing Compliance (HUD / LIHTC / TRACS / iMAX) | React / TypeScript / JavaScript | Python / FastAPI / Django | Java / Spring Boot | PHP / Laravel | PostgreSQL / MySQL / Oracle | AWS (S3, Lambda, RDS, EC2) | REST APIs / SOA | Test-Driven Development | CI/CD (GitHub Actions) | Docker / Kubernetes | Agile / Scrum / Kanban | Code Review & Mentoring | Data Pipeline & ETL | PHI Security & Compliance

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- Virginia, USA (Hybrid)

- Build Python/FastAPI backend services and React/TypeScript frontends on AWS, shipped through GitHub Actions CI/CD with Docker containerization.
- Integrate LLM capabilities (Anthropic Claude) into financial data workflows with structured outputs, eval harnesses, and guardrails for sensitive data handling.
- Build ML pipelines (scikit-learn, XGBoost) from feature engineering through prediction APIs; instrument services with Prometheus and OpenTelemetry.

Entrata -- Affordable Housing & Property Management SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant SaaS platform, 100K+ monthly active users

- Became the **technical domain expert in Affordable Housing compliance** -- LIHTC certification workflows, HUD 202D guidelines, TRACS/iMAX submission pipelines, income eligibility calculations, and HAP/Tax Credit reporting across thousands of properties.
- Owned the annual **HUD income-limit import end-to-end for 6 years** -- transforming federal data into ~180,000 domain records with stored procedure logic, validation, effective-date handling, and rollback preparation. Zero compliance violations across 6 annual cycles.
- Built and maintained **LIHTC certification and recertification workflows** -- complex data gathering, multi-agency submission logic, and recurring compliance deadlines within a diverse regulatory environment.
- Created **56 state-specific housing certification documents** via automated merge-field pipelines, eliminating manual document preparation for property managers across all states.
- Delivered **20+ features end-to-end** (Python, PHP, Java, React, PostgreSQL) across 120+ Agile sprints -- from initial scoping and design through release and production support.
- Built RabbitMQ-based async processing for rent updates across thousands of tenant records; optimized 200+ SQL queries (30% performance gain); implemented PHI encryption (Libsodium) covering 100% of resident data.
- Reviewed **300+ pull requests**, mentored junior engineers, and contributed to shared engineering standards and TDD practices.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built affordable housing compliance modules (Python, PHP, Java, PostgreSQL) for LIHTC, Section 8, and HUD rental workflows; delivered reconciliation pipelines processing 1.6M HUD/Tax Credit records.
- Led AWS S3 cloud migration from legacy server storage for Affordable Housing module.

PROJECTS

Leasel -- Affordable Housing Compliance Platform

Python · React · FastAPI · PostgreSQL · LangGraph · TDD

Full-stack compliance platform for affordable housing operations -- lease document ingestion, clause extraction, compliance Q&A with cited answers, and human escalation for regulated decisions. Built with TDD: LLM-as-judge CI harness across 1,240 test cases (93.6% precision, 87.9% F1). Role-based access control ensures eligibility and financial decisions always route to staff.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- LLM Evaluation Toolkit

Python · FastAPI · GitHub Actions CI/CD · TDD

15-endpoint FastAPI backend with automated testing via golden datasets in CI. End-to-end regression suite across five LLM providers; scores outputs on accuracy, hallucination, retrieval, completeness, and compliance.

github.com/venkatalakshmiparimala/AgentProbe

CrashClarity -- Road Accident ML Analysis

Python · scikit-learn · React · GitHub Pages

ML pipeline on 104K+ UK accident records (Random Forest, Logistic Regression, 85%+ accuracy) with interactive React dashboard -- demonstrates full-stack delivery from data layer through frontend.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017